

# SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

**Course Code:** 23CS002PC304

**Course Title:** AI Assisted Coding

**Year/Sem:** III / II

**Assignment Number:** 3.1 / 24

Name: O.Narasimha Raju

HT No: 2303A53034

Batch: 46

## Lab Objectives

- Understand different prompt engineering techniques
- Use Zero-shot, One-shot, and Few-shot prompting
- Generate correct code using AI prompts
- Analyze effectiveness of different prompt styles

## Task 1: Zero-Shot Prompting

**Prompt:** Write a Python program to check whether a number is a palindrome.

**Python Code**

```
num = input("Enter number: ")

if num == num[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

**Explanation:** The problem is solved without giving any example to AI.

## Task 2: One-Shot Prompting

**Prompt:** Example:  $5! = 120$  Now write a Python program to calculate factorial.

**Python Code**

```
n = int(input("Enter number: "))
fact = 1

for i in range(1, n+1):
    fact *= i

print("Factorial:", fact)
```

**Explanation:** One example helps AI generate correct logic.

## Task 3: Few-Shot Prompting

**Prompt:** 153 → Armstrong 370 → Armstrong Write a program to check Armstrong number.

**Python Code**

```
num = int(input("Enter number: "))
temp = num
sum = 0
digits = len(str(num))

while temp > 0:
    digit = temp % 10
    sum += digit ** digits
    temp //= 10

if sum == num:
    print("Armstrong")
else:
    print("Not Armstrong")
```

**Explanation:** Multiple examples improve AI accuracy.

## Task 4: Prompt Comparison

- Zero-shot is fast but less accurate

- One-shot improves correctness
- Few-shot gives best results
- More context leads to better AI responses

## Conclusion

Prompt engineering plays a crucial role in AI-assisted coding.

Providing examples and clear instructions improves accuracy and efficiency of generated code.